FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

2

RULE – BASED AGENT

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I. BACKGROUND INFORMATION

A Rule-Based Agent is a type of intelligent agent that operates by following predefined rules to decide its actions. In this lab, students will work with a vacuum cleaner agent that can clean rooms or move between rooms depending on their state (dirty or clean).

The goal is to understand how agents perceive the environment, act according to rules, and update the environment.

II. TECHNICAL ASSESSMENT

RULE – BASED AGENT

Objective

At the end of this laboratory activity, students are expected to:

1. Explain the concept of rule-based agents.
2. Implement a rule-based agent in Python.
3. Interact with an environment by perceiving and acting based on rules.
4. Observe and analyze the behavior of an agent in a controlled environment.

Materials Needed

- Computer with Python 3.x installed
- Text editor or IDE (e.g., VSCode, PyCharm)
- Optional: Jupyter Notebook for interactive testing

General Instructions

1. This **Technical Assessment (TA4)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA2_lastname.pdf**
   (Example: TA2_Patilano.pdf)
8. Submit the **python file** and **PDF** file.

Laboratory Task

Pre-Lab Preparation

1. Review the Python concepts:
   - Classes and objects
   - If-Else statements

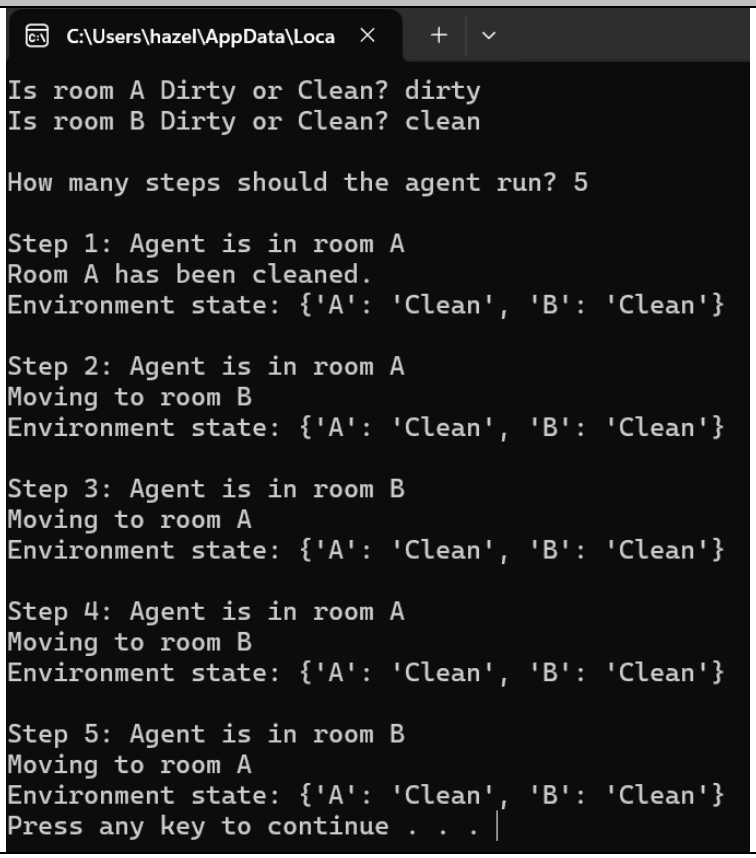
- Loops (for loop)
- Functions / methods

2. Understand the environment:
   - Two rooms: A and B
   - Each room can be Dirty or Clean

Task 1: Set Up the Environment

1. Initialize two rooms using a Python dictionary.
2. Allow the user to set the **initial state** of each room (dirty or clean).
3. Write a function to **check if a room is dirty** and a function to **clean a room**.

Sample Output



Task 2: Implement the Rule-Based Agent

1. Create a class for the agent.
2. Implement the **perceive-and-act method**:
   - Rule 1: If the current room is dirty, clean it.
   - Rule 2: If the current room is clean, move to the other room.
3. Implement a **move method** to switch rooms.

Task 3: Run the Simulation

1. Ask the user how many steps the simulation should run.
2. Run the agent in a loop for the specified steps.

3. After each step, print:
    - The agent's current location
    - The action taken (cleaned/moved)
    - Current state of the environment

SOURCE CODE:

```
# Function to check if the current room is dirty
def is_dirty(environment, room):
    return environment[room] == "Dirty"

# Function to clean the current room
def clean_room(environment, room):
    environment[room] = "Clean"

# Get the initial condition of Room A and Room B
room_a = input("Is room A Dirty or Clean? ").capitalize()
room_b = input("Is room B Dirty or Clean? ").capitalize()

# Create the environment using a dictionary
environment = {
    "A": room_a,
    "B": room_b
}

print()

class Agent:
    def __init__(self):
        # The agent starts in Room A
        self.current_room = "A"

    # Method to move to the other room
    def move(self):
        if self.current_room == "A":
            self.current_room = "B"
        else:
            self.current_room = "A"

    # Perceive and act based on rules
    def perceive_and_act(self, environment):

        # Rule 1: If the current room is dirty, clean it
        if is_dirty(environment, self.current_room):
            clean_room(environment, self.current_room)
```

```
            return "Room " + self.current_room + " has been
cleaned."

         # Rule 2: If the current room is clean, move to the
other room
        else:
            old_room = self.current_room
            self.move()
            return "Moving to room " + self.current_room

# Ask the user how many steps the agent should run
steps = int(input("How many steps should the agent run? "))

print()

# Create the agent
agent = Agent()

# Run the simulation
for step in range(1, steps + 1):

         print("Step",   step,   ":   Agent   is   in   room",
agent.current_room)

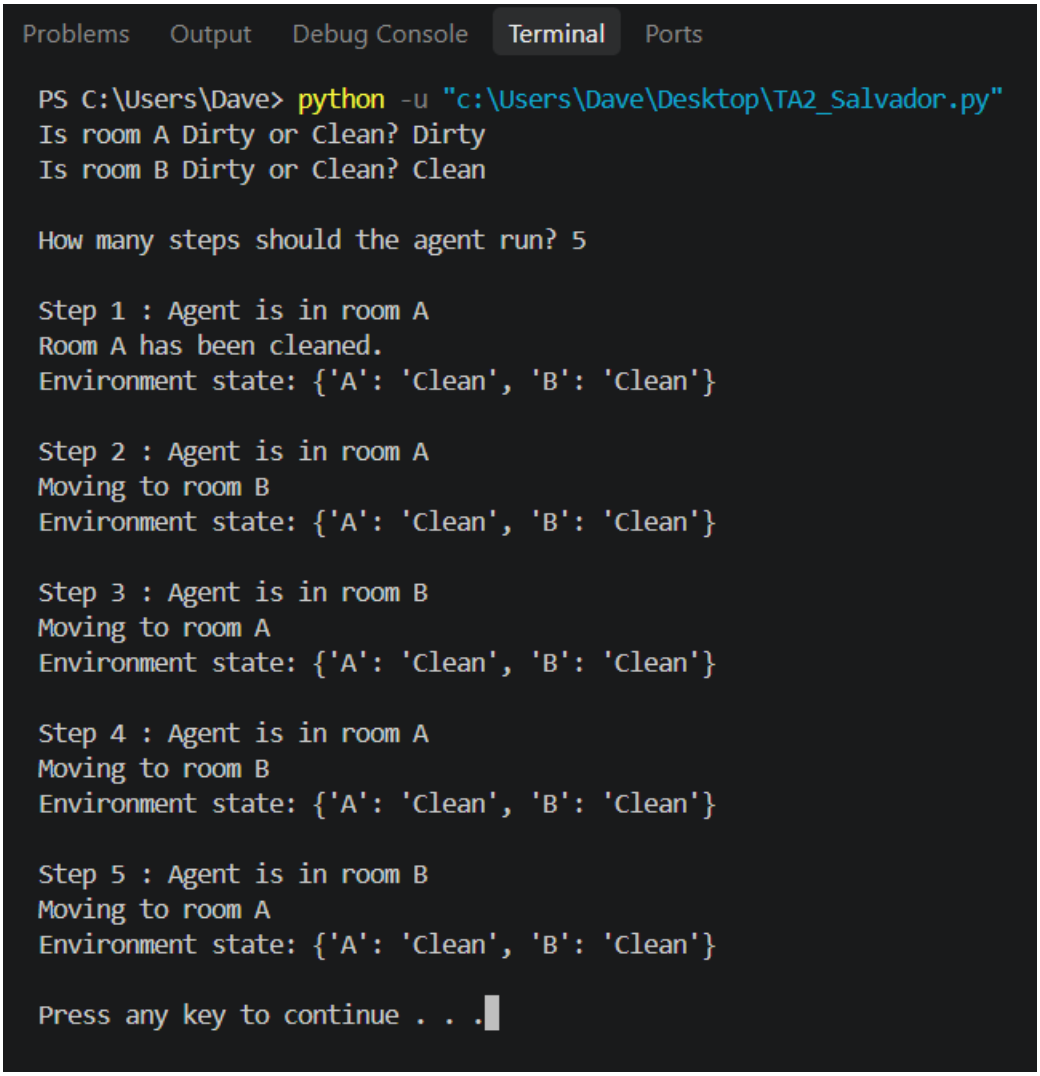
    # Agent perceives the environment and performs an action
    action = agent.perceive_and_act(environment)

    # Display the action
    print(action)

    # Display the current environment state
    print("Environment state:", environment)
    print()

input("Press any key to continue . . .")
```

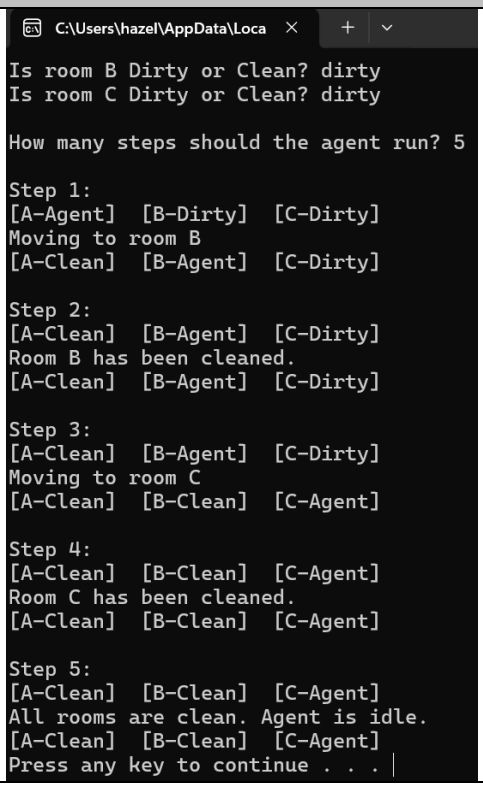
OUTPUT:


Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\Dave> python -u "c:\Users\Dave\Desktop\TA2_Salvador.py"
Is room A Dirty or Clean? Dirty
Is room B Dirty or Clean? Clean
How many steps should the agent run? 5
Step 1 : Agent is in room A
Room A has been cleaned.
Environment state: {'A': 'Clean', 'B': 'Clean'}
Step 2 : Agent is in room A
Moving to room B
Environment state: {'A': 'Clean', 'B': 'Clean'}
Step 3 : Agent is in room B
Moving to room A
Environment state: {'A': 'Clean', 'B': 'Clean'}
Step 4 : Agent is in room A
Moving to room B
Environment state: {'A': 'Clean', 'B': 'Clean'}
Step 5 : Agent is in room B
Moving to room A
Environment state: {'A': 'Clean', 'B': 'Clean'}
Press any key to continue . . .

Optional (Bonus Task)

- Add a **third room (C)** and update the agent logic.
- Make the agent **randomly choose which dirty room to clean next**.
- Visualize the agent moving between rooms using a **text-based grid**.

Sample Output (Bonus Task)



Your Code - Laboratory Tasks ⇩

SOURCE CODE:

```
import random

# THREE-ROOM CLEANING AGENT

# Ask the user for the condition of each room
rooms = {}

for room in ["A", "B", "C"]:
      condition = input(f"Is room {room} Dirty or Clean?
").capitalize()
```

```
    while condition not in ["Dirty", "Clean"]:
        condition = input(f"Please enter Dirty or Clean for room
{room}: ").capitalize()

    rooms[room] = condition

# Agent starts in Room A
agent_position = "A"

# Display the current state of the rooms
def display_rooms():
    for room in ["A", "B", "C"]:
        if room == agent_position:
            print(f"[{room}-Agent]", end="  ")
        else:
            print(f"[{room}-{rooms[room]}]", end="  ")
    print()

# Ask how many steps the agent should run
steps = int(input("\nHow many steps should the agent run? "))

print("\nInitial State:")
display_rooms()

# Agent starts moving
for step in range(1, steps + 1):

    print(f"\nStep {step}:")
    display_rooms()

    # Find all dirty rooms
     dirty_rooms = [room for room in rooms if rooms[room] ==
"Dirty"]

    # Stop if all rooms are clean
    if not dirty_rooms:
        print("All rooms are clean. Agent is idle.")
        break

    # If the current room is dirty, clean it
    if rooms[agent_position] == "Dirty":
        rooms[agent_position] = "Clean"
        print(f"Room {agent_position} has been cleaned.")

    else:
```

```
        # Randomly choose another dirty room
         dirty_rooms = [room for room in rooms if rooms[room] ==
"Dirty"]

        if dirty_rooms:
            next_room = random.choice(dirty_rooms)
            print(f"Moving to room {next_room}")
            agent_position = next_room

    print("Current State:")
    display_rooms()

# Final result
print("\nFinal State:")
display_rooms()

if all(rooms[room] == "Clean" for room in rooms):
    print("All rooms are clean. Agent is idle.")
else:
    print("Agent stopped because the step limit was reached.")
```

OUTPUT:

```
PS C:\Users\Dave> python -u "c:\Users\Dave\Desktop\Bonus Task.py
Is room A Dirty or Clean? Dirty
Is room B Dirty or Clean? Clean
Please enter Dirty or Clean for room B: Clean
Is room C Dirty or Clean? Dirty

How many steps should the agent run? 3

Initial State:
[A-Agent]  [B-Clean]  [C-Dirty]

Step 1:
[A-Agent]  [B-Clean]  [C-Dirty]
Room A has been cleaned.
Current State:
[A-Agent]  [B-Clean]  [C-Dirty]

Step 2:
[A-Agent]  [B-Clean]  [C-Dirty]
Moving to room C
Current State:
[A-Clean]  [B-Clean]  [C-Agent]

Step 3:
[A-Clean]  [B-Clean]  [C-Agent]
Room C has been cleaned.
Current State:
[A-Clean]  [B-Clean]  [C-Agent]

Final State:
[A-Clean]  [B-Clean]  [C-Agent]
All rooms are clean. Agent is idle.
PS C:\Users\Dave>
```

Task 4: Analysis Questions

After running the simulation, answer the following:

1. What rules does the agent follow?
2. How does the agent decide when to clean and when to move?
3. What happens if both rooms are already clean at the start?
4. How would you modify the agent to handle **more than two rooms**?

Answer the Questions ⇩

1. **The agent follows two simple rules. First, if the room it is currently in is dirty, it will clean the room. Second, if the room is already clean, it will move to the other room. These rules help the agent decide what action to take based on the condition of the room.**
2. **The agent checks the condition of the room where it is currently located. If the room is dirty, the agent cleans it because cleaning is the priority. If the room is already clean, there is nothing to clean, so the agent moves to the other room to check it.**
3. **If both rooms are already clean, the agent will not need to clean anything. It will simply move from one room to the other every step because both rooms are already clean. The environment will stay clean throughout the simulation.**
4. **To handle more than two rooms, I would change the program so that it can store several rooms in a list or dictionary. Instead of only switching between Room A and Room B, the agent could keep track of its current room and move to the next available room. It could check each room one by one and clean it whenever it finds a dirty room.**